

HYDRONEXIS - AI · ACCOUNTING MODULE · PURCHASING SUB-MODULE

ACC - PUR - 003

Purchase Request

Entry point for every purchasing workflow

MODULE	HydroNexis-AI · Accounting · Purchasing · Screen 002
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Purpose

This screen creates every purchase request in HydroNexis. Nothing can be purchased without an approved Purchase Request. Every Purchase Request receives a unique PR Number and becomes fully traceable through the workflow to a closed PO.

Access matrix

ROLE	CREATE	EDIT	APPROVE	CANCEL
Employee	Yes	Draft only	No	Draft only
Department Manager	Yes	Yes	Level 1	Yes
Purchasing Officer	Yes	Yes	No	Yes
Purchasing Manager	Yes	Yes	Level 2	Yes
Accounting Manager	Yes	Yes	Financial	Yes
CEO	Yes	Yes	Final	Yes

PR Number

Automatically generated on Save Draft. Format:

PR-YYYY -NNNNNN

Example: PR-2026-000458 . Immutable once assigned. Sequence is monotonic per calendar year; the counter resets on 1 January.

Header fields (read-only after creation)

- Purchase Request No.
- Status
- Created By
- Created Date
- Last Modified
- Revision
- Priority
- Workflow Stage

Workflow

Linear pipeline. Each stage advances only when its exit conditions are met.

#	STAGE	EXIT CONDITION
01	Draft	User clicks Submit and all Submit-Validation checks pass
02	Submitted	AI Validation job completes
03	AI Validation	No blocking issues, or user acknowledges warnings
04	Department Approval	Level 1 approver decision recorded
05	Purchasing Review	Purchasing Officer/Manager assigns handling
06	Inventory Verification	SOS reconfirms need after any pending transfers
07	Supplier Sourcing	Purchasing selects supplier list or triggers Supplier Discovery
08	RFQ	RFQ sent; awaiting responses
09	Quotation Evaluation	At least one qualifying quote received and scored
10	Approval	All required approval levels recorded
11	Purchase Order	PO created and sent
12	Receiving	Goods received against PO
13	Completed	PO closed, three-way match complete

Main form — general information

FIELD	TYPE	RULE	REQUIRED
Request Title	Text	Max 150 chars	Yes
Department	Dropdown	Sourced from Departments	Yes
Project	Dropdown	Sourced from Projects	No
Cost Center	Dropdown	Accounting integration	Yes
Budget Code	Lookup	Accounting Budget	Yes
Required Delivery Date	Date	Must be ≥ today; may be constrained by category	Yes
Priority	Enum	Normal · High · Urgent · Emergency	Yes
Justification	Text area	Min 30 · Max 3000 chars	Yes

Item grid

A single Purchase Request may contain multiple items. Every row is independently validated.

COLUMN	RULE
Line No.	Auto, 1..N per PR
Item Code	Free text OR lookup into SOS Items · triggers Intelligent Item Recognition
Item Description	Populated from lookup, editable
Category	Inherited from item; overridable
Quantity	Positive number · decimals allowed per unit definition
Unit	From item master
Estimated Unit Price	Populated from last purchase · editable
Estimated Total	Computed: qty × unit price · read-only
Currency	Defaults to company currency · overridable per line
Preferred Brand	Optional; drives supplier suggestion ranking
Alternative Brand Allowed	Boolean; if false, Nexi must not suggest substitutes
Warehouse Destination	Which warehouse receives the goods
Required Date	Per-line override of the header Required Delivery Date
Remarks	Free text
Attachment	Per-line file upload (drawing, spec sheet, etc.)

Intelligent item recognition

When an Item Code is entered, Nexi immediately queries every source she has and returns results inline within 500 ms:

- SOS Inventory (on-hand + reserved)
- Previous purchases of this item
- Supplier database (who has sold this item)
- Warehouse availability (all sites)

- Historical pricing (last 12 months, per supplier)
- Open purchase orders for the same item
- Approved brands
- Existing draft/submitted duplicate requests

AI suggestion panel

Right-side rail, always visible while the form is open. Suggestions appear as cards; each card is dismissible and time-stamped. Representative outputs:

- Current stock available at Warehouse A — internal transfer recommended
- Transfer from GH5 Warehouse (48 units on hand)
- Supplier already delivering tomorrow — add to that PO
- Previous price ₱142 · current market estimate ₱138
- Preferred supplier: ABC Industrial (score 87, 8 past orders)
- Savings opportunity: ₱3,250 by consolidating with PR-000452
- Risk level: LOW

Attachments

RULE	VALUE
Formats	PDF · DWG · DXF · Excel · Word · Images (JPG/PNG/HEIC) · Videos (MP4/MOV) · ZIP
Max size per file	500 MB
Virus scan	Every upload scanned before persistence; failed scans quarantined and audit-logged
Retention	Tied to the PR record; kept per company retention policy (default 7 years)

Technical drawing preview

Built-in viewer for PDF/DWG/DXF/Image line-item attachments. Features: zoom · rotate · fullscreen · version history · markup (annotations saved back to the PR).

Budget validation

Real-time. Every quantity or unit-price change re-runs the check.

- Budget available for the selected Budget Code
- Remaining budget in the current period
- Project budget (if a project is selected)
- Department budget
- Monthly allocation
- Annual allocation

RESULT	BEHAVIOUR
OK	Green tick shown next to the amount
Insufficient	Yellow warning — submission allowed with manager acknowledgement
Exceeded	Red block — Submit button disabled until fixed or a higher-level override is granted

Inventory validation

Calls SOS Inventory on every recomputation.

- Current stock
- Reserved stock
- Incoming orders

- Warehouse transfers in flight
- Alternative items (same category / substitutable)

RULE

If inventory already exists that satisfies the request, Nexi recommends internal transfer instead of purchasing and marks the PR line as blocked pending user acknowledgement.

Duplicate detection

Searches for open or submitted PRs matching all of:

- Same Item
- Same Department
- Same Project
- Same week
- Same Supplier (if preferred is set)
- Same Quantity band ($\pm 20\%$)

If matches are found, the user is shown a "Possible duplicate" panel with links and must explicitly confirm before submission.

Supplier intelligence preview

Per line item, shows:

- Preferred supplier
- Last supplier
- Last purchase price
- Average lead time
- Supplier rating (0-100)
- Warranty history

- Quality rating

AI recommendation panel

Persistent right-side panel. Recommendations are informational unless a policy flag makes them blocking. Example outputs:

- Purchase not required — internal transfer covers the need
- Internal transfer recommended (GH5 → GH1, 48 units)
- Combine with PR-458 to unlock 5 % volume discount
- Use existing contract with Supplier XYZ (contract expires 2026-11)
- Wait two days — this item price is trending down 8 %
- Alternative supplier available with 30 % shorter lead time
- Risk alert — supplier under credit review

Approval matrix preview

Shown to the user before Submit. Reflects the live approval matrix based on total value, department, category, and any category-specific overrides.

Example route:

Department Manager → Purchasing Manager → Accounting Manager → CEO

Route is snapshotted onto the PR at submission time; subsequent matrix changes do not retro-affect in-flight requests.

Submit validation

Submit is disabled unless ALL of the following are true:

- All mandatory fields complete
- Budget check is OK or warning-acknowledged
- Every line item has a valid quantity, unit, and estimated total
- Required attachments (per category rules) are uploaded and scanned
- AI Validation job has returned
- Every raised duplicate warning has been acknowledged

Buttons

BUTTON	EFFECT
Save Draft	Persists as Draft; PR Number assigned on first save
Submit	Advances to AI Validation stage; kicks off the workflow
Validate	Runs the full validation suite without submitting
Check Inventory	Forces a fresh SOS query and re-evaluates transfer suggestions
Find Supplier	Opens 6.1 AI Supplier Search seeded with the current item(s)
Estimate Cost	Refreshes unit-price estimates from historical + market data
Cancel	Cancels the PR (respects Access matrix)
Print	Print-friendly view
Export PDF	Downloads a PDF of the PR with a QR link back
Clone Request	Duplicates the PR into a new Draft

Notifications

On submission, notifications are dispatched immediately to:

- Department Manager (Level 1 approver)

- Purchasing Officer + Manager
- Accounting (if a Budget Code override is used)
- AI Engine (for the validation job)

Audit trail

Every change to a Purchase Request is persisted and cannot be deleted.

FIELD	CAPTURED
User	User id + display name at time of change
Timestamp	UTC, ms precision
Old Value	JSON snapshot
New Value	JSON snapshot
Reason	Free-text, required for edits after Submitted stage
IP Address	Source IP
Device	User-agent fingerprint

APIs

METHOD + PATH	PURPOSE
POST /api/purchasing/pr	Create a new PR (Draft)
PUT /api/purchasing/pr/{id}	Update fields or line items
GET /api/purchasing/pr/{id}	Fetch a single PR with items + workflow state
POST /api/purchasing/pr/{id}/submit	Submit the PR — advances to AI Validation stage
POST /api/purchasing/pr/{id}/cancel	Cancel the PR
POST /api/purchasing/pr/{id}/clone	Clone into a new Draft
GET /api/sos/items	Item lookup for the Item Code field
GET /api/sos/stock	Current + reserved stock for a given item
GET /api/budget/check	Real-time budget validation
GET /api/projects	Project dropdown source
GET /api/departments	Department dropdown source
GET /api/suppliers/history	Supplier intelligence panel
POST /api/attachments/upload	Multi-part upload; returns attachment ids
POST /api/ai/pr/validate	Run AI validation for the PR

Database tables

- PurchaseRequest
- PurchaseRequestItems

- PurchaseRequestWorkflow
- PurchaseRequestAttachments
- Departments
- Projects
- Budget
- Inventory
- Suppliers
- AuditLog
- Notifications

Acceptance criteria

- PR number is generated automatically and is immutable.
- A single PR supports N line items with independent, per-line validation.
- Budget checks and SOS inventory checks are real-time and re-run on every relevant edit.
- Duplicate requests are surfaced before submission and require explicit acknowledgement.
- AI recommendations are non-blocking by default; policy flags can make specific recommendations blocking.
- All changes are captured by the audit trail with the fields listed above.
- Submission creates the workflow instance and notifies the next approver within 5 seconds.
- A submitted PR cannot be edited except through the versioned Revision mechanism.